

Mingli Sun

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1. EDUCATION

MFA in Arts and Design, Advanced Environmental Design Sep 2022 - Jun 2025
GPA: 4.25/5; College of Design and Innovation (D&I), **Tongji University**, Shanghai

BEng in Industrial Design Sep 2016 - Jun 2020
GPA: 3.69/4; Mechanical and Electrical Engineering School, **Beijing Information Science & Technology University (BISTU)**, Beijing

2. ACADEMIC EXPERIENCE

Research Assistant Jun 2025 – Jun 2026, 1 year
College of Architecture and Urban Planning (CAUP), Tongji University

- ♦ Led on-site technical verification and optimization for a National Key R&D Program-funded project, with a peer-reviewed Chinese journal publication.
- ♦ Mentored 2 undergraduate research initiatives, with a peer-reviewed conference publication.

Research Intern Nov 2024 – Jan 2025, 3 months
Institute of Future Human Habitats, Tsinghua Shenzhen International Graduate School

- ♦ Developed an end-to-end digital workflow integrating structural design and performance analysis, with a peer-reviewed conference publication.

Technical Instructor Feb 2023 – July 2023, 6 months
FabLab O Shanghai (Affiliated with the MIT FabLab Global Network)

- ♦ Provided technical guidance for p5.js in STEAM courses and contributed to the AI for Kids curriculum directed by Prof. Kostas Terzidis.

3. PUBLICATIONS

- ♦ Sun, M. & Chai, H. (2026). Intelligent Robotic Sanding System for Building Exterior Walls Integrated with Residential Building Manufacturing Machines. *Housing Science*, 2026(6), 80–84. <https://doi.org/10.3969/j.issn.1002-0454.2026.06.011> (Chinese journal article)
- ♦ Sun, M., Zhou, Y., Liu, Y., Zhu, Y., Yang, Q., Liu, Y., & Chai, H. (2026). Graph-Based Generative Design of Mortise-and-Tenon Joints in Timber Structures. *Proceedings of the 8th International Conference on Computational Design and Robotic Fabrication (CDRF2026)*. (Accepted; EI)
- ♦ Sun, M., Chen, J., & Terzidis, K. (2025). Text-guided Fine-tuning Diffusion Model Enhanced the Spatial Envision of “Modular” Structures: Personalized Diffusion Model in the Conceptual Design of Architectural Structures. *Proceedings of the 30th CAADRIA Conference: Architectural Informatics (CAADRIA 2025)*, Vol. 1, 49–58. <https://doi.org/10.52842/conf.caadria.2025.1.049> (EI)
- ♦ Sun, M., Terzidis, K., & Chen, J. (2024). Facilitation or Inhibition: The Influence of Generative Artificial Intelligence on Design Reasoning for Modular Structures. *Proceedings of the IASS 2024 Symposium: Redefining the Art of Structural Design*, 109. <https://doi.org/10.3929/ethz-b-000728788> (EI)
- ♦ Sun, M., Buš, P., Chen, J., & Sun, K. (2024). Co-Intelligent Design Toward Modular Structures. *Proceedings of the 29th CAADRIA Conference: Accelerated Design (CAADRIA 2024)*, Vol.3, 171–180. <https://doi.org/10.52842/conf.caadria.2024.3.171> (EI)

4. RESEARCH EXPERIENCE

3.1 Architectural Design

Graph Learning-Based Generative Design of Mortise-and-Tenon Timber Joints

Sep 2025 – Jun 2026

Fields: Artificial Neural Networks, Algorithm Design, Timber Architecture

Technology: Graph Neural Networks (GNNs), Python, PyTorch, PyG, Rhino - Grasshopper

Leveraging GNNs to capture the geometric, topological, and mechanical features of mortise-and-tenon joints, this project transforms the experienced-based craftsmanship into a digital and intelligent-driven paradigm.

1. Participated in collaborative discussions to establish the research framework, construct the joint database with 200+ samples, and define graph representations.
2. **Developed** a data processing procedure to convert 3D joint samples into graph data, conducting data inspection and augmentation expanding the dataset to 1,500+; **formulated** evaluation metrics and calculation approaches for joint performance analysis; **developed** a procedure to generate model inputs based on user-defined assembly sequences and installation directions.
3. Contributed to model testing and optimization, and **primarily authored** the manuscript published by CDRF2026.

Generative AI Images for Early Conceptual Design: A Creativity Support Index (CSI) Study

Feb 2024 – Jun 2025

Fields: Generative AI, Human-AI Interaction, Design Ethics

Technology: Stable Diffusion (SD), LoRA, Python, PyTorch

This study investigates how generative AI images influence early conceptual design, employing the Creativity Support Index (CSI) and verbal protocol analysis to study human-AI interaction.

1. **Proposed and led** the research project, including defining research questions, planning technical route, and designing experimental protocols.
2. **Developed** a customized diffusion model via LoRA fine-tuning, deeply participating in dataset preparation, training configuration, model evaluation.
3. **Designed** a user-centric empirical study integrating verbal protocol analysis and quantitative metrics. **Co-authored** a peer-reviewed conference paper.

An Imitation Learning-Based Design Agent for Modular Assembly-Oriented Structures

Jun 2023 – Sep 2023

Fields: Human-AI Interaction, Imitation Learning, Collaborative Design

Technology: Generative Adversarial Imitation Learning (GAIL), Unity3D, ML-Agents Toolkit

Trains an agent through Generative Adversarial Imitation Learning (GAIL) and Proximal Policy Optimization (PPO) in Unity3D with ML-Agents Toolkit, using human demonstration trajectories to learn design reasoning strategies for modular assembly-oriented structures.

1. **Developed** a Unity3D-based interactive environment for agent training, including environment and object scripting, parameter configuration, and interaction mechanism implementation.
2. Implemented agent training using GAIL within the ML-Agents Toolkit, including demonstration collection, parameter tuning, iterative training and evaluation.
3. Refined AI-generated outcomes considering manufacturing and assembly requirements. **Co-authored** a peer-reviewed conference publication.

3.2 Construction Engineering

Intelligent Robotic System for High-Rise Building Façade Finishing Jul 2025 – Jun 2026

Funded by National Key R&D Program of China | Fields: Construction Robotics, Algorithm Design
Contributes to the later-stage validation, parametrically models operation and environment, defines evaluation metrics for operational precision, and validates performance through field testing.

- 1. Led on-site experiments, established evaluation metrics for robotic system performance, and proposed hardware control optimization strategies validated to improve operational accuracy.
- 2. Published a peer-reviewed journal article in Chinese and co-invented a patent.

3.3 Arts and Design

Urban Park Digital Art Driven by Eco-Acoustic Index Visualization Sep. 2023 – Dec. 2023

Fields: Data Visualization, Communication Design | Technology: Tableau, TouchDesigner, Arduino
Designed a digital art experience driven by real-time eco-acoustic index visualization to support ecological education and public awareness of urban bird habitats.

- 1. Collaborated with environmental science researchers in interdisciplinary workshops and field research, and developed visual encoding strategies for eco-acoustic data.
- 2. Independently designed and implemented the installation through TouchDesigner-Arduino integration. The project outcome was published in a peer-reviewed design conference CUMULUS.

5. HONORS, AWARDS & GRANTS

Shanghai Outstanding Graduate Award (top 5%)	Shanghai Municipal Education Commission	2025
Xiangcheng High-Tech Talent Scholarship (endowed, top ~1%)	Tongji University	2024
International Academic Conference Travel Grant	Tongji University	2024
International and Hong Kong, Macao, and Taiwan Exchange Grant	Tongji University	2023
Beijing Outstanding Graduate Award (top 5%)	Beijing Municipal Education Commission	2020

6. PROFESSIONAL SKILLS

- ♦ Machine Learning Methods: Graph Neural Networks, Imitation Learning, Diffusion Models
- ♦ Programming Languages & Frameworks: Python, C#, PyTorch, PyTorch Geometric
- ♦ Computational Geometry: Algorithm Development, Parametric Modeling (Rhino, Grasshopper)
- ♦ Digital Fabrication & Construction: 3D Printing, CNC Fabrication, Construction Robotics
- ♦ Empirical Research Methods: Field Research, Interviews, Persona Development, Think-Aloud

*Languages: Mandarin Chinese (Native), English (Professional Working Proficiency)

7. REFERENCES

Prof. Kostas Terzidis
Professor, College of Design and Innovation (D&I), Tongji University, Shanghai, China
Email: kostas@tongji.edu.cn
Relationship: Master's Supervisor

Prof. Hua Chai
Associate Professor, College of Architecture and Urban Planning (CAUP), Tongji University, Shanghai, China
Email: chaihua@tongji.edu.cn
Relationship: Research Advisor